

Chief Digital and Artificial Intelligence Office

Approach to Globally Unique Identifiers

“Data-driven decision-making requires DoD data to be linked such that relationships and dependencies can be uncovered and maintained.” – 2020 DoD Data Strategy

EXECUTIVE SUMMARY

The fourth goal of Department of Defense’s (DoD) Data Strategy is to Make Data Linked, with a key indicator of progress being the implementation of “globally unique identifiers so data can be easily discovered, linked, retrieved, and referenced,” as stated in the Strategy. We must be able to manage data as it transits across the enterprise, having confidence in origin (provenance) and modifications over time (lineage). Accordingly, while unique identifiers (UIDs) are essential to making data linked, they are also key considerations as DoD organizations make their data visible and trustworthy.

Well-formed UIDs in DoD must be:

- Unique and consistent across the DoD enterprise, through adherence to a known technical pattern
- Persistent for all time, inactivating versus deleting UIDs to enable longitudinal analytics
- Non-Information-Bearing, no additional content beyond the unique identifier
- Supportive of most legacy systems, limiting UIDs to 32-bits
- Human- and machine-readable, enable direct machine use as well as human understanding

The CDAO will pursue achieving the new approach by:

- Prototype capability – ‘learn by doing’ – for an enterprise UID service
- Provide guidance on UID patterns and processes for piloting and future incorporation into DoD Issuances
- Engaging major UID programs to drive alignment to the enterprise approach proposed herein

Progress will be demonstrated when use of priority data products is accelerated due to UID generated with a common approach by multiple organizations. This plan assumes a rapid pace for assessing current needs and capabilities, a regular drumbeat of agile development, and measured progress towards enterprise scale with specific goals for mission outcomes.

BACKGROUND

Current State: Non-aligned UID implementations. DoDI 8320.03, *Unique Identification (UID) Standards for Supporting the DoD Information Enterprise*, identifies a number of unique issuing authorities, as presented in Figure 1, placing responsibility on functional leaders, without technical guidance to ensure a consistent or interoperable approach. Currently the Department does not have a process in place to align the independent UIDs that are issued from each of the authorities (Figure 1). The multitude of *Associated Unique Identifiers* (Figure 1) standards and patterns that are used by each authority further complicates the process of collecting all data assets IDs into a single registry that is easily searchable.

Table. DoD UID Standards

DoD UID Standard Entity	Associated Unique Identifier
Personnel Identification	The DoD Identification Number is the common name for the Electronic Data Interchange Personal Identifier (EDI-PI), which is structured in accordance with DoDI 1000.30 (Reference (g)).
Real Property and Real Property Sites	The real property unique identifier (RPUID) and real property site unique identifier (RPSUID) are structured in accordance with DoDI 4165.14 (Reference (h)).
Global Force Management	The Global Force Management Data Initiative is structured to include the organization unique identifier (OUID) for internal DoD use in accordance with DoDI 8260.03 (Reference (i)), DoD Manual 8260.03, Volume 1 (Reference (j)), and DoD Manual 8263.03, Volume 2 (Reference (k)).
Tangible Personal Property	Item unique identification (IUID) is structured in accordance with DoDI 8320.04 (Reference (l)).
External DoD Business Partners	External organization unique identification (EOUID) for external DoD mission partners that conduct business with DoD under contracts or other agreements is determined in accordance with DoDI 8320.06 (Reference (m)).
Unit Reference Numbers (URNs)	URNs are associated with variable message formats in accordance with CJCS Instruction 3156.01A (Reference (n)).
Transportation Tracking Number (TTN)	TTNs are associated with the marking and labeling of shipment units for unit cargo movements in accordance with Part III of Defense Transportation Regulation 4500.9-R (Reference (o)).

Figure 1 - UID Authorities

Implication: Challenges in assembling trustworthy data across functional domains. These inconsistencies undercut efficacy in the enterprise and DoD's ability to consume data from any source with a certainty of identity. This multi-UID approach creates obstacles to assemble data, impeding analytic engines' ability to easily and clearly identify specific assets when: computing the trade-space for decision making, enumerating what is contributing to the analysis, and determining changes to be made. An enterprise of DoD's size must approach UIDs in a way that ensures data assets approach 100% certainty of their individual identity. UIDs are vital to all operational and analytical pursuits of the Department, including inventory (logistics), personnel (P&R), and decision support systems (advanced analytics including Artificial Intelligence (AI) disciplines), where each requires full confidence the data is unique, discrete, and non-duplicative to enable accurate data-driven business and mission decisions. There are also mission benefits to single-source UIDs by eliminating the time and energy to cross-correlate disparate UIDs when consuming multiple-authority data assets. The ability to assemble and analyze vast amounts of data is one of the foundational needs for training and predicting outcomes for machine learning. Multiple UID regimes and formats cause massive engineering efforts to attempt unifying UID equivalents and harmonizing data prior to use, and there is no guarantee that the process will identify all duplications.

Collisions of UIDs from disparate authorities may result in errors of understanding and intent of data and require manual intervention. Established analytics processes cannot overcome these challenges at scale, with confidence and accuracy, and incur significant risk to data-driven decision-making. Additionally, without well-formed UIDs it is difficult to track and associate data assets and the amplifying metadata from multiple communities without pre-coordination.

Implication: Inefficiencies in data lifecycle management – determining uniqueness and consistency at scale. The lack of mapping, non-uniform UID structures, and potential for multiple UIDs assigned to a single data asset impacts the ability to effectively identify and manage discrete data objects across multiple authorities and impedes the ability to cross-reference data objects at scale.

As the Department matures its competencies in data management and catalogs are virtually unified, it is critical to know if a data set in one catalog is the same within another catalog. If independent authorities are managing catalogs using differing UID standards and patterns, there is a significant risk that if the same data asset are presented in each catalog there would be two distinct UIDs based on the local authority process, and the consumer would not have any indication that the data sets were identical.

Tracking every asset (e.g.: person, place, materiel, or software), virtually, through linked data, is practically untenable if multiple UID models are used. While software and other mapping and merging techniques can be created, these approaches require additional resources and time to establish an automated aligning process and result in computational overhead each time any of the disparate UID assets are brought together. In the current state, it is likely that one data asset could have multiple UIDs if it has more than one purpose across the UID authorities.

Scaling and accuracy for an enterprise requires a reliable means of discreetly and uniquely identifying digital assets without inconsistency. The globally unique ID concept allows for the creation of digital assets without contradiction (same ID for two different things or two IDs for one thing), the linking of separately recorded pieces of information (updating or enriching data), and the precise removal or masking of links, so that for example, health information (e.g., HIPAA 45 CFR §164.514) is not individually identifiable unless anonymized. Through proper UID management, attributes for data can be used to determine access (sharing and safeguarding) across organizations, business practices and mission areas.

STRATEGIC APPROACH

Overcoming the challenge of scale requires a globally unique identifier that is persistent for all time and strictly associates to one and only one asset. The most basic UID model would be to have a single source generated UID that is able to track every object / asset for the foreseeable future. Single source generation, however, is not feasible given DoD's decentralized and segmented architecture.

DoD also needs to accommodate legacy systems that are typically unable to handle a number larger than 32-bits in size, ruling out a unifying approach that anchors on industry standards, such as Universal UID (UUID) (July 2005 RFC 4122)) or Global UID (Microsoft's equivalent UUID implementation) both of which use larger numbers (128-bits in size).

For this discussion an asset is any single item that is distinctly identifiable, which may include, but is not limited to: person (actually a physical object), physical object (e.g.: vehicle, desk), logical object (e.g.: software, organization) or any other discrete object/asset that the Department needs to discretely track.

A more adaptable solution that considers the DoD legacy systems, borrows from the Global Force Management Data Initiative (GFM DI) model that uses a pair of two 32-bit numbers, a centrally managed seed and an extension managed by the local organization, and then appends the two numbers together (the concept of tuples).

The approach proposed herein uses a more general structure to enable longer 32-bit tuple groups where each echelon, as determined by the echelon's parent organization, uses cascaded 32-bit identifiers which are locally issued and managed. This allows for “tuples” (stringing the numbers together as prepended identifiers) so that within an organizations echelon hierarchy there is a maximum of 2^{32} UIDs, over 4.3 billion numbers available. UIDs will grow in length as their associated data asset is shared with external organizations, necessitating prepended identifiers to ensure global uniqueness (adding the parents' strings to the local UID). The model continues to support legacy systems that have a physical constraint of handling numbers that cannot exceed 32-bits, while ensure that shared data assets can be exposed to enterprise consumers with assurance of unique identity.

The model (presented immediately following this text) shows how a parent organization (DoD) issues a UID to an immediate child organization (US Army) and then the US Army issues the next level of UID to the Regular US Army, keeping approximately 4.3 billion UIDs for each subgroup within each level.

UID is a sequential number within each Echelon (from $1-2^{32}$) [~ 4.3 billion]

DoD – 4.3 Billion

USA – 4.3 billion

USA-Regular 4.3 Billion

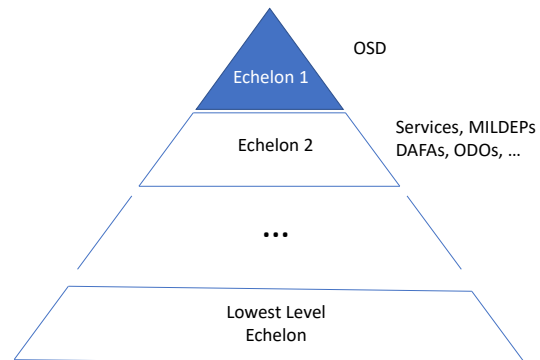
USA-Reserve 4.3 Billion

USA-Guard 4.3 Billion

DON – 4.3 Billion

DLA – 4.3 Billion

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For example:

DoD: UID [0]

USA: UID = [1]-[0]

USA Regular: UID = [1]-[1]-[0]

{Note: these next two are from the same parent Organization}

USA-Regular 2nd Tier Organization (first one): UID = [1]-[1]-[1]-[1] *{No children}*

USA-Regular 2nd Tier Organization (56,000th one): UID = [1]-[1]-[1]-[56,001]-[0]

USA Reserve: UID = [1]-[2]-[0]

USA Guard NY: UID = [1]-[3]-[0]

USA Guard MN: UID = [1]-[4]-[0]

DON: UID = [2]-[0]

DLA: UID = [3]-[0]

In this example the DoD issued USA the UID of “1”. The USA then issued the Regular USA (versus the Reserve and Guard) a UID of “1”. Note that these “1”s are not the same UID; as the tuples are UID-UID-UID resulting in [0]-[1]-[1], where DoD is [0], USA is the first [1], and finally the Regular USA is the second [1]. Typically the first [0] is dropped, so that the Regular USA would be [1]-[1]. There is no parent to DoD in the enterprise and the [0] is assumed.

Additionally, the trailing [0] indicates that this organization is intended to issue UIDs to subordinate organizations. So the USA [1]-[0] indicates that the parent organization (DoD) has issued the child organization (USA) the UID of [1] and recognizes that USA will be issuing UIDs to subordinates with the trailing [0].

The use of the “-[0]” at the end of a UID string provides an indicator that this is a “parent” organization and has children organizations. The elegance of this approach is that the data systems can manage the local UIDs that are assigned (the last [x] number in the tuple) and are only required to prepend the entire parent string when it is sending data beyond its own organization. The system can also prevent an echelon level organization from issuing UIDs to a child organization if there is no [0] at the end of their UID.

The parent string is always the same fixed string and is computationally inexpensive adding near-zero overhead and provides full object / asset tracking across the Department for the foreseeable future. This allows for tuple prepending any time data is shared beyond the local echelon.

For example, if the Regular USA were to pass data to the Reserve USA the tuple would be [1]-[1]-[object UID] where the first [1] is USA, the second [1] is Regular USA and the [object UID] is the “data asset” unique identifier assigned by the Regular USA. This allows the Reserve USA to know that this asset is coming from the Regular USA by querying the UID software service. The same number also works for the DON as the [1]-[1]-[object UID] would be unchanged even when sent outside of the USA.

This approach also allows a data system to know the organizational origin of a data asset based on the fully constructed UID, providing a provenance attribute and lineage tracking opportunity. External data asset contributions are also accommodated by this approach. When a data object / asset is “inbound” from an external entity, the DoD organization that is taking receipt simply acquires a UID and affixes it to the data object / asset. Once this is accomplished, the DoD can start to analyze the number of duplicates and multiple payments that have been issued for the same external data.

TECHNICAL REQUIREMENTS:

The establishment of a UID across the enterprise will require both technical guidance and policy issuances. The technical requirements for a singularly unique identifier (resulting in globally unique UIDs) are few and straight forward in nature. The use of a UID pattern will allow for

continuous innovation in implementation and long-term flexibility in how numbers are instantiated with any UID issuing organization. The identifier must minimally be:

- Unique across the entire enterprise
- Persistent for all time [Inactivating versus deleting allows for longitudinal analytics]
- Non-information bearing [obtaining the number provides no additional information]
- Usable by most legacy systems [32-bit numbers are typically acceptable]
- Supports distributed services model
- Near-instantaneously issuance (numbers are generated on-demand)
- Consistent across all functional disciplines
- Both Human and Machine readable

The initial steps to achieve the new UID model include:

- Staffing the UID model for community awareness
- Issue a memorandum to align enterprise behavior for UID generation and use
- Adjusting DODI 8320.03 to use a single format and continuous UID issuance
- Employing governance to adjudicate any specific challenges and issues
- Provide oversight to support the enterprise adoption and deployment
- Implement measures and metrics to provide insights into adoption progress

ENTERPRISE CAPABILITY:

The DoD adoption of a global UID program (which has a consistent unified pattern), as shown in Figure 2, enables the Department to achieve a better understanding of its data assets and enables the advanced analytics necessary to maintain our advantage.

The proposed model is also extensible over time as the legacy barriers are reduced (larger numbers can be handled), each UID tuple can be expanded to larger number options (e.g., 2 x 32-bit tuples, 4x 32-bit tuples, etc.) without any impact to the current UID set. This is possible as the numbers are issued sequentially in order and increasing the size of the UID space simply allows for larger numbers, which follow in-line to the already issued UIDs. This provides for a near-infinite identity space for all time.

The UIDs are also foundational to supporting other Enterprise Data Management (EDM) services that are natural consumers of UIDs to optimize and accelerate processes. Some examples include advanced analytics on collections of data sets (UIDs can be used to “grab” these data assets / objects with certainty of uniqueness), creation of profiles and Bills of Material (BOMs), which use UIDs to identify the assets required to compose assemblies and sub-assemblies or tracking very large contracts (F-35) that have requirements resulting in millions of unique assets for delivery. There is no limit to how the UIDs are applied in this scalable model.

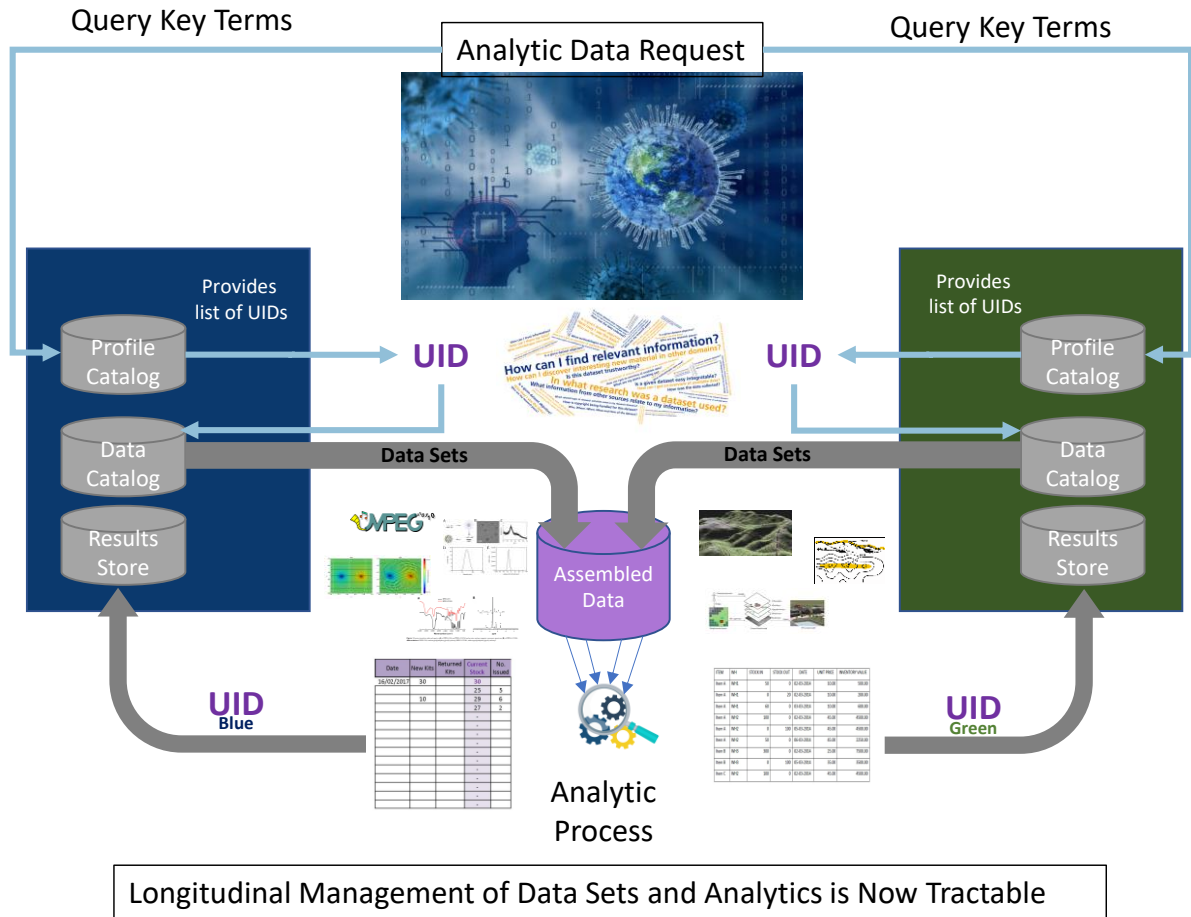


Figure 2 - UID Enterprise Model 1